VALIDATION PLAN
OPERATIONAL QUALIFICATION
SHELF LIFE & DEGRADATION ANALYSIS MODULE

JR Validated Environment — Shelf Life & Degradation Analysis Module v1.0

Field	Value
Document Number	JR-VP-SHELF-001
Title	Validation Plan — Operational Qualification, Shelf Life & Degradation Analysis Module
System	JR Validated Environment — Shelf Life Module
Module Version	1.0
Document Version	1.0
Status	Approved
Effective Date	2026-04-19
Author	Joep Rous
Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-04-19	Joep Rous	Initial release. Covers all 5 Shelf Life scripts in module v1.0. 57 test cases including --transform log (TC-SHELF-LIN-013..016).

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-04-19
Reviewer	[Name]		
Approver (QA)	[Name]		

1. Purpose

This document defines the Validation Plan for the Operational Qualification (OQ) of the Shelf Life & Degradation Analysis module included in version 1.0 of the JR Validated Environment. It specifies the user requirements, test cases, acceptance criteria, and traceability between requirements and tests needed to demonstrate that each Shelf Life script performs correctly for its intended use.

Execution of the tests defined in this plan, together with satisfactory results and documented evidence, constitutes the OQ for the Shelf Life module.

2. Scope

2.1 In Scope

- All 5 R scripts in repos/shelf_life/R/ (jrc_shelf_life_q10, jrc_shelf_life_arrhenius, jrc_shelf_life_linear, jrc_shelf_life_poolability, jrc_shelf_life_extrapolate)
- Correct computation of acceleration factor (AF) and real-time equivalent via Q10 method (ASTM F1980) and Arrhenius kinetics (ISO 11607 / ICH Q1E)
- Correct linear degradation modelling: Brown-Forsythe homogeneity test, OLS regression, shelf life estimation as intersection of CI of predicted mean with specification limit
- Correct --transform log behaviour: log-scale OLS with CI back-transformed via exp() to original measurement scale
- Correct ICH Q1E batch poolability analysis: two-step ANCOVA at α=0.25, FULL POOL / PARTIAL POOL / DO NOT POOL decision
- Correct extrapolation of CI from a saved model CSV: ICH Q1E 50% warning and 100% hard stop; transform field read and applied correctly
- Correct rejection of invalid inputs with informative error messages
- Correct file I/O behaviour (CSV input, PNG and model CSV output to ~/Downloads/)
- Verification that all R scripts enforce execution within the jrrun validated environment by checking RENV_PATHS_ROOT at startup

2.2 Out of Scope

- Infrastructure scripts in bin/ and admin/ (covered by IQ; see JR-VP-001)
- Performance qualification (PQ) — real product stability studies are conducted separately under individual stability protocols
- User acceptance testing (UAT)
- Validation of R language interpreter and third-party packages (ggplot2, nlme) — package integrity is confirmed at installation via SHA256 (JR-IQ-001)
- Statistical validation of the underlying methods (Arrhenius kinetics, ICH Q1E poolability decision rules, OLS inference) — method references are cited in Section 3

3. Regulatory Basis and References

3.1 Regulations

Reference	Title

21 CFR Part 820, §820.70(i)	Quality System Regulation — Automated Data Processing
21 CFR Part 820, §820.75	Process Validation
21 CFR Part 11	Electronic Records; Electronic Signatures
ISO 13485:2016, §7.5.6	Validation of processes for production and service provision

3.2 FDA Guidance

Reference	Title	Date
FDA GPSV	General Principles of Software Validation; Final Guidance for Industry and FDA Staff	January 2002
FDA CSA	Computer Software Assurance for Production and Quality System Software; Draft Guidance	September 2022

3.3 Standards and Methods

Reference	Title
ASTM F1980-16	Standard Guide for Accelerated Aging of Sterile Barrier Systems for Medical Devices. Q10 method basis. AF = Q10^(ΔT/10).
ISO 11607-1:2019	Packaging for terminally sterilized medical devices — Part 1: Requirements for materials, sterile barrier systems and packaging systems. References Arrhenius method for accelerated ageing.
ICH Q1E	Evaluation for Stability Data. International Council for Harmonisation. Poolability decision rules (two-step ANCOVA at α=0.25); extrapolation limits (50%/100% beyond last observation).
ICH Q1A(R2)	Stability Testing of New Drug Substances and Products. Defines shelf life estimation from stability regression and CI of the mean.
ISO 13485:2016	Medical devices — Quality management systems — Requirements for regulatory purposes.
Arrhenius (1889)	Arrhenius SA. Über die Reaktionsgeschwindigkeit bei der Inversion von Rohrzucker durch Säuren. Zeitschrift für physikalische Chemie. AF = exp(Ea/R × (1/T_real − 1/T_accel)); R = 1.987×10^{-3} kcal/(mol·K).
Brown & Forsythe (1974)	Robust tests for equality of variances. Journal of the American Statistical Association, 69(346), 364–367. Implemented as one-way ANOVA on absolute deviations from group medians using base R lm() and anova() — equivalent to car::leveneTest(..., center=median).

3.4 Internal Documents

Document Number	Title
JR-VP-001	Validation Plan — Installation Qualification, JR Validated Environment v1.0.0
JR-IQ-001	IQ Execution Evidence (docs/IQ_validation_*.txt)
JR-VP-002	Validation Plan — OQ, Community Script Suite

4. Definitions and Abbreviations

Term	Definition
AF	Acceleration Factor — ratio of real-time to accelerated ageing duration that achieves equivalent degradation.
Q10	Temperature coefficient — the factor by which reaction rate increases per 10°C rise. Typically 1.5–3.0 for medical device degradation.
Ea	Activation energy (kcal/mol) — Arrhenius model parameter. Typical range for medical device degradation: 10–25 kcal/mol.
R (gas constant)	1.987×10^{-3} kcal/(mol·K) — used in the Arrhenius equation.
ICH Q1E	International Council for Harmonisation guideline on evaluation for stability data. Defines poolability analysis, shelf life estimation, and extrapolation limits.
ANCOVA	Analysis of Covariance. Used in ICH Q1E poolability: tests batch×time interaction (Step 1) and batch main effect (Step 2) at α=0.25.

Brown-Forsythe	Homogeneity-of-variance test robust to non-normal data. Implemented as one-way ANOVA on absolute deviations from group medians (base R only — no car package).
OLS	Ordinary Least Squares regression. Used in jrc_shelf_life_linear to fit value ~ time (or log(value) ~ time with --transform log).
CI of the mean	Confidence interval for the predicted mean response. Shelf life is estimated as the time at which the CI bound of the predicted mean crosses the specification limit (ICH Q1E approach, not prediction interval).
Transform (log)	--transform log flag in jrc_shelf_life_linear. Fits lm(log(value) ~ time); CI is back-transformed via exp() to the original measurement scale. Use for right-skewed data or log-normal degradation.
FULL POOL	ICH Q1E poolability decision: slopes and intercepts of all batches are not statistically different (Step 1 $p > 0.25$ and Step 2 $p > 0.25$). A single regression line may be used for all batches.
PARTIAL POOL	ICH Q1E poolability decision: slopes are not statistically different (Step 1 $p > 0.25$) but intercepts differ (Step 2 $p \leq 0.25$). Use the batch with the shortest shelf life.
DO NOT POOL	ICH Q1E poolability decision: slopes differ significantly (Step 1 $p \leq 0.25$). Separate regression lines must be used; the batch with the shortest shelf life applies.
IQ	Installation Qualification
OQ	Operational Qualification
PQ	Performance Qualification
UR	User Requirement
TC	Test Case
jrrun	The validated bash entry-point script that sets the controlled environment before executing any script
CSV	Comma-separated values input file
PNG	Portable Network Graphics output file; plots saved to ~/Downloads/

5. User Requirements

The following user requirements define the expected behaviour of the Shelf Life & Degradation Analysis module. Each requirement is verifiable by the test cases in Section 7.

UR ID	Title	Requirement
UR-SHELF-001	Bypass protection — RENV_PATHS_ROOT	All Shelf Life scripts shall check the RENV_PATHS_ROOT environment variable at startup and call stop() with an informative message if it is not set, preventing execution outside the validated jrrun environment.
UR-SHELF-002	Error handling — non-zero exit	All Shelf Life scripts shall exit with a non-zero status code whenever an error condition is detected, and shall print an informative message.
UR-SHELF-003	Q10 — acceleration factor computation	jrc_shelf_life_q10 shall compute the acceleration factor as AF = Q10^((accel_temp − real_temp) / 10) and the real-time equivalent as real_time = accel_time × AF (ASTM F1980).
UR-SHELF-004	Q10 — sensitivity table	jrc_shelf_life_q10 shall report a sensitivity table showing AF and real-time equivalent for Q10 − 0.5, Q10 (stated), and Q10 + 0.5.
UR-SHELF-005	Q10 — range warning	jrc_shelf_life_q10 shall emit a warning when Q10 is outside the typical range [1.5, 3.0] and continue with exit 0.
UR-SHELF-006	Q10 — temperature direction check	jrc_shelf_life_q10 shall reject inputs where accel_temp ≤ real_temp with a non-zero exit, because no acceleration occurs.
UR-SHELF-007	Arrhenius — acceleration factor computation	jrc_shelf_life_arrhenius shall compute AF = exp(Ea/R × (1/T_real − 1/T_accel)) where R = 1.987×10^{-3} kcal/(mol·K) and temperatures are in Kelvin (ISO 11607 / ICH Q1E).
UR-SHELF-008	Arrhenius — unit flag	jrc_shelf_life_arrhenius shall accept --unit C (Celsius, default) or --unit K (Kelvin) and produce identical AF values when equivalent temperatures are supplied in either unit.
UR-SHELF-009	Arrhenius — sensitivity table	jrc_shelf_life_arrhenius shall report a sensitivity table showing AF and real-time equivalent for Ea − 2, Ea (stated), and Ea + 2 kcal/mol.
UR-SHELF-010	Arrhenius — Ea range warning	jrc_shelf_life_arrhenius shall emit a warning when Ea is outside the

		typical range [10, 25] kcal/mol and continue with exit 0.
UR-SHELF-011	Arrhenius — temperature direction check	jrc_shelf_life_arrhenius shall reject inputs where accel_temp ≤ real_temp or an invalid --unit value is supplied, with a non-zero exit.
UR-SHELF-012	Linear — CSV input	jrc_shelf_life_linear shall accept a CSV file with headers 'time' and 'value' containing individual measurements (not aggregated means). At least 3 distinct time points are required.
UR-SHELF-013	Linear — Brown-Forsythe homogeneity test	jrc_shelf_life_linear shall perform the Brown-Forsythe test for homogeneity of variance across time points (base R implementation: ANOVA on absolute deviations from group medians). The result is reported; heterogeneity is a warning, not an error.
UR-SHELF-014	Linear — OLS regression and shelf life estimation	jrc_shelf_life_linear shall fit lm(value ~ time) by OLS and solve for the shelf life as the time at which the lower (or upper, if --direction high) confidence bound of the predicted mean crosses the specification limit.
UR-SHELF-015	Linear — --direction flag	jrc_shelf_life_linear shall accept --direction low (default, value must stay above spec) and --direction high (value must stay below spec, growing impurity use case).
UR-SHELF-016	Linear — --transform log flag	jrc_shelf_life_linear shall accept --transform log, which fits lm(log(value) ~ time). The confidence interval of the predicted mean is back-transformed via exp() before comparing to the specification limit. All values must be strictly positive. The model CSV shall record the transform field as 'log'.
UR-SHELF-017	Linear — PNG output	jrc_shelf_life_linear shall save a plot PNG to ~/Downloads/ with a timestamp prefix and the suffix _jrc_shelf_life_linear.png.
UR-SHELF-018	Linear — model CSV output	jrc_shelf_life_linear shall save a model coefficient CSV to ~/Downloads/ containing intercept, slope, residual SE, n, t_bar, Sxx, last_time, spec_limit, confidence, direction, transform, and source_file. This file is consumed by jrc_shelf_life_extrapolate.
UR-SHELF-019	Linear — reject values below spec at t=0	jrc_shelf_life_linear shall reject datasets where the observed mean or fit is already below the specification limit at the first time point (direction=low), with a non-zero exit.
UR-SHELF-020	Poolability — CSV input	jrc_shelf_life_poolability shall accept a CSV file with headers 'batch', 'time', and 'value'. At least 2 distinct batches and at least 3 distinct time points per batch are required.
UR-SHELF-021	Poolability — ICH Q1E two-step ANCOVA	jrc_shelf_life_poolability shall perform the ICH Q1E two-step ANCOVA at α=0.25: Step 1 tests the batch×time interaction (slope equality); Step 2 tests the batch main effect (intercept equality). Both F-statistics and p-values are reported.
UR-SHELF-022	Poolability — pooling decision	jrc_shelf_life_poolability shall report one of three decisions: FULL POOL (Step 1 and Step 2 both p > 0.25), PARTIAL POOL (Step 1 p > 0.25, Step 2 p ≤ 0.25), DO NOT POOL (Step 1 p ≤ 0.25).
UR-SHELF-023	Poolability — PNG output	jrc_shelf_life_poolability shall save a batch degradation plot PNG to ~/Downloads/ with a timestamp prefix and the suffix _jrc_shelf_life_poolability.png.
UR-SHELF-024	Extrapolate — model CSV input and verification	jrc_shelf_life_extrapolate shall read a model CSV produced by jrc_shelf_life_linear and verify that the 'script' field identifies jrc_shelf_life_linear. Non-existent or wrong-script files shall cause a non-zero exit.
UR-SHELF-025	Extrapolate — CI projection with transform support	jrc_shelf_life_extrapolate shall compute the fitted value and confidence interval of the mean at the target time using the stored model coefficients. If the model CSV contains transform=log, the CI shall be back-transformed via exp() before display and comparison to the specification limit.
UR-SHELF-026	Extrapolate — ICH Q1E extrapolation limits	jrc_shelf_life_extrapolate shall emit a warning when the target time is more than 50% beyond the last observation, and shall reject (non-zero exit) when the target time is more than 100% beyond the last observation.

UR-SHELF-027	Extrapolate — input validation	jrc_shelf_life_extrapolate shall reject a negative target_time with a non-zero exit.

6. Test Environment

Component	Specification
Operating System	macOS 15.x (Apple Silicon) or Windows 10/11 (Git Bash)
R Version	As installed per JR-IQ-001 (R 4.3.x or later)
Python Version	As installed per JR-IQ-001 (Python 3.11.x or later)
Test Framework	pytest 8.x
Test runner	repos/shelf_life/admin_shelf_life_oq
Test file	repos/shelf_life/oq/test_shelf_life_suite.py (57 tests)
Test data	repos/shelf_life/oq/data/ (13 CSV fixtures)
OQ venv	~/.venvs/<PROJECT_ID>_oq/ (shared with other module OQ suites)
Output directory	~/Downloads/ (PNG and model CSV verified by mtime glob)
Entry point	bin/jrrun (integrity-checked execution)

7. Test Cases

All test cases are implemented in repos/shelf_life/oq/test_shelf_life_suite.py. Test data files reside in repos/shelf_life/oq/data/. Numerical reference values for correctness tests are computed independently of the R scripts using pure Python arithmetic (math.exp(), OLS in Python, bisection) and NIST t-table values.

7.1 jrc_shelf_life_q10

TC-SHELF-Q10-001

Tests: UR-SHELF-003
Command: `jrrun jrc_shelf_life_q10.R 2.0 55 25 26`
Rationale: Valid inputs.
Pass criterion: exit 0; 'Acceleration factor', 'Real-time equivalent', 'Sensitivity' in output

TC-SHELF-Q10-002

Tests: UR-SHELF-003
Command: `jrrun jrc_shelf_life_q10.R 2.0 55 25 26`
Rationale: Known values: AF = 2.0^(30/10) = 8.0 (exact); real_time = 26 × 8.0 = 208.0 (exact). Reference computed by integer arithmetic, independent of R.
Pass criterion: exit 0; AF = 8.0 ± 0.001; real_time = 208.0 ± 0.01

TC-SHELF-Q10-003

Tests: UR-SHELF-004
Command: `jrrun jrc_shelf_life_q10.R 2.0 55 25 26`
Rationale: Sensitivity table rows for Q10=1.5, Q10=2.0 (stated), Q10=2.5.
Pass criterion: exit 0; '1.5', 'stated value', '2.5' in output

TC-SHELF-Q10-004

Tests: UR-SHELF-005
Command: `jrrun jrc_shelf_life_q10.R 1.2 55 25 26`
Rationale: Q10=1.2 (below [1.5, 3.0] typical range).
Pass criterion: exit 0; range warning in output

TC-SHELF-Q10-005

Tests: UR-SHELF-006
Command: `jrrun jrc_shelf_life_q10.R 2.0 25 55 26`

Rationale: accel_temp (25) ≤ real_temp (55) — no acceleration.
Pass criterion: exit non-zero

TC-SHELF-Q10-006

Tests: UR-SHELF-002
Command: `jrrun jrc_shelf_life_q10.R  (no arguments)`
Rationale: Missing arguments.
Pass criterion: exit non-zero or 'Usage' in output

TC-SHELF-Q10-007

Tests: UR-SHELF-002
Command: `jrrun jrc_shelf_life_q10.R abc 55 25 26`
Rationale: Non-numeric q10.
Pass criterion: exit non-zero

TC-SHELF-Q10-008

Tests: UR-SHELF-002
Command: `jrrun jrc_shelf_life_q10.R 2.0 55 25 -5`
Rationale: Negative accel_time.
Pass criterion: exit non-zero

TC-SHELF-Q10-009

Tests: UR-SHELF-002
Command: `jrrun jrc_shelf_life_q10.R --help`
Rationale: --help flag.
Pass criterion: exit 0; 'Usage' in output

TC-SHELF-Q10-010

Tests: UR-SHELF-001
Command: `Rscript --vanilla repos/shelf_life/R/jrc_shelf_life_q10.R 2.0 55 25 26 (RENV_PATHS_ROOT unset)`
Rationale: Direct Rscript call bypasses jrrun.
Pass criterion: exit non-zero; 'RENV_PATHS_ROOT' in output

7.2 jrc_shelf_life_arrhenius

TC-SHELF-ARR-001

Tests: UR-SHELF-007
Command: `jrrun jrc_shelf_life_arrhenius.R 55 25 17.0 26`
Rationale: Valid inputs.
Pass criterion: exit 0; 'Acceleration factor', 'Real-time equivalent', 'Sensitivity' in output

TC-SHELF-ARR-002

Tests: UR-SHELF-007
Command: `jrrun jrc_shelf_life_arrhenius.R 55 25 17.0 26`
Rationale: Known values: AF = exp(17.0/1.987e-3 × (1/298.15 − 1/328.15)) = 13.7826. Reference computed via math.exp() in pure Python, independent of R. Tolerance ±0.001.
Pass criterion: exit 0; AF = 13.7826 ± 0.001; real_time = 358.35 ± 0.05

TC-SHELF-ARR-003

Tests: UR-SHELF-008
Command: `jrrun jrc_shelf_life_arrhenius.R 328.15 298.15 17.0 26 --unit K`
Rationale: Kelvin inputs equivalent to 55°C/25°C Celsius run.
Pass criterion: exit 0; AF matches Celsius run ± 0.01

TC-SHELF-ARR-004

Tests: UR-SHELF-009
Command: `jrrun jrc_shelf_life_arrhenius.R 55 25 17.0 26`
Rationale: Sensitivity table rows for Ea=15.0, Ea=17.0 (stated), Ea=19.0.
Pass criterion: exit 0; '15.0', '17.0', '19.0' in output

TC-SHELF-ARR-005

Tests: UR-SHELF-010
Command: `jrrun jrc_shelf_life_arrhenius.R 55 25 30.0 26`
Rationale: Ea=30.0 (above [10, 25] typical range).
Pass criterion: exit 0; range warning in output

TC-SHELF-ARR-006

Tests: UR-SHELF-011
Command: `jrrun jrc_shelf_life_arrhenius.R 25 55 17.0 26`
Rationale: accel_temp (25) ≤ real_temp (55).
Pass criterion: exit non-zero

TC-SHELF-ARR-007

Tests: UR-SHELF-002
Command: `jrrun jrc_shelf_life_arrhenius.R  (no arguments)`
Rationale: Missing arguments.
Pass criterion: exit non-zero or 'Usage' in output

TC-SHELF-ARR-008

Tests: UR-SHELF-011
Command: `jrrun jrc_shelf_life_arrhenius.R 55 25 17.0 26 --unit F`
Rationale: Invalid --unit value (Fahrenheit not supported).
Pass criterion: exit non-zero

TC-SHELF-ARR-009

Tests: UR-SHELF-002
Command: `jrrun jrc_shelf_life_arrhenius.R --help`
Rationale: --help flag.
Pass criterion: exit 0; 'Usage' in output

TC-SHELF-ARR-010

Tests: UR-SHELF-001
Command: `Rscript --vanilla repos/shelf_life/R/jrc_shelf_life_arrhenius.R 55 25 17.0 26 (RENV_PATHS_ROOT unset)`
Rationale: Direct Rscript call bypasses jrrun.
Pass criterion: exit non-zero; 'RENV_PATHS_ROOT' in output

7.3 jrc_shelf_life_linear

TC-SHELF-LIN-001

Tests: UR-SHELF-012, UR-SHELF-013, UR-SHELF-014
Command: `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95`
Rationale: Valid 15-row homogeneous dataset (5 time points × 3 units).
Pass criterion: exit 0; 'Homogeneity of Variance', 'Regression', 'Shelf Life Estimate' in output

TC-SHELF-LIN-002

Tests: UR-SHELF-014

Command: `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95`

Rationale: Known dataset, spec=80, conf=0.95. Reference shelf life computed by pure-Python OLS + bisection: b0=100.647, b1=−0.807, sigma=0.686 (hand-verified). t_crit(0.975, df=13)=2.16037 (NIST t-table). Expected shelf life ≈24.723. Tolerance ±0.05.

Pass criterion: exit 0; shelf life = 24.723 ± 0.05

TC-SHELF-LIN-003

Tests: UR-SHELF-013

Command: `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95`

Rationale: Homogeneous variance dataset.

Pass criterion: exit 0; 'homogeneous' or variance pass in output

TC-SHELF-LIN-004

Tests: UR-SHELF-013

Command: `jrrun jrc_shelf_life_linear.R shelf_life_linear_heterogeneous.csv 75.0 0.95`

Rationale: Heterogeneous variance dataset.

Pass criterion: exit 0; heterogeneous variance warning in output

TC-SHELF-LIN-005

Tests: UR-SHELF-017

Command: `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95`

Rationale: PNG output.

Pass criterion: exit 0; *_jrc_shelf_life_linear.png found in ~/Downloads/

TC-SHELF-LIN-006

Tests: UR-SHELF-018

Command: `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95`

Rationale: Model CSV output.

Pass criterion: exit 0; *_jrc_shelf_life_linear_model.csv found in ~/Downloads/

TC-SHELF-LIN-007

Tests: UR-SHELF-015

Command: `jrrun jrc_shelf_life_linear.R shelf_life_linear_direction_high.csv 0.20 0.95 --direction high`

Rationale: --direction high (growing impurity data).

Pass criterion: exit 0; 'upper' or 'high' in output

TC-SHELF-LIN-008

Tests: UR-SHELF-019

Command: `jrrun jrc_shelf_life_linear.R shelf_life_linear_below_spec.csv 80.0 0.95`

Rationale: Values already below spec at t=0 (direction=low).

Pass criterion: exit non-zero

TC-SHELF-LIN-009

Tests: UR-SHELF-002

Command: `jrrun jrc_shelf_life_linear.R  (no arguments)`

Rationale: Missing arguments.

Pass criterion: exit non-zero or 'Usage' in output

TC-SHELF-LIN-010

Tests: UR-SHELF-012
Command: `jrrun jrc_shelf_life_linear.R shelf_life_linear_missing_col.csv 80.0 0.95`
Rationale: CSV missing 'value' column.
Pass criterion: exit non-zero; 'value' in error output

TC-SHELF-LIN-011

Tests: UR-SHELF-012
Command: `jrrun jrc_shelf_life_linear.R shelf_life_linear_few_timepoints.csv 90.0 0.95`
Rationale: Only 2 distinct time points (< 3 required).
Pass criterion: exit non-zero

TC-SHELF-LIN-012

Tests: UR-SHELF-001
Command: `Rscript --vanilla repos/shelf_life/R/jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95 (RENV_PATHS_ROOT unset)`
Rationale: Direct Rscript call bypasses jrrun.
Pass criterion: exit non-zero; 'RENV_PATHS_ROOT' in output

TC-SHELF-LIN-013

Tests: UR-SHELF-016
Command: `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95 --transform log`
Rationale: --transform log flag; log transform noted in output.
Pass criterion: exit 0; 'log' in output; 'Shelf Life Estimate' in output

TC-SHELF-LIN-014

Tests: UR-SHELF-016
Command: `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95 --transform log`
Rationale: Known log-linear shelf life: OLS on log(value) ~ time computed in pure Python. CI back-transformed via exp(). Reference shelf life computed via bisection. Tolerance ±0.05.
Pass criterion: exit 0; shelf life = _SHELF_LIFE_LOG_EXPECTED ± 0.05

TC-SHELF-LIN-015

Tests: UR-SHELF-016
Command: `jrrun jrc_shelf_life_linear.R <data with value=0> 70.0 0.95 --transform log`
Rationale: Data contains value ≤ 0 with --transform log.
Pass criterion: exit non-zero; 'positive' or '<= 0' in error output

TC-SHELF-LIN-016

Tests: UR-SHELF-016, UR-SHELF-018
Command: `jrrun jrc_shelf_life_linear.R shelf_life_linear_homogeneous.csv 80.0 0.95 --transform log`
Rationale: Model CSV must record transform field.
Pass criterion: exit 0; model CSV contains 'transform' = 'log'

7.4 jrc_shelf_life_poolability

TC-SHELF-POOL-001

Tests: UR-SHELF-020, UR-SHELF-021, UR-SHELF-022
Command: `jrrun jrc_shelf_life_poolability.R shelf_life_pool_poolable.csv`

Rationale: Valid 3-batch dataset.
Pass criterion: exit 0; 'Decision', 'POOL', 'Step 1', 'Step 2' in output

TC-SHELF-POOL-002

Tests: UR-SHELF-022
Command: `jrrun jrc_shelf_life_poolability.R shelf_life_pool_poolable.csv`
Rationale: Dataset with equal slopes and intercepts. Both ANCOVA steps p > 0.25.
Pass criterion: exit 0; 'FULL POOL' in output; Step 1 p > 0.25; Step 2 p > 0.25

TC-SHELF-POOL-003

Tests: UR-SHELF-022
Command: `jrrun jrc_shelf_life_poolability.R shelf_life_pool_no_pool.csv`
Rationale: Dataset with clearly different slopes (A=−0.4, B=−0.8, C=−1.2 per unit time). Step 1 interaction must be highly significant (p < 0.05).
Pass criterion: exit 0; 'DO NOT POOL' in output; Step 1 p < 0.05

TC-SHELF-POOL-004

Tests: UR-SHELF-022
Command: `jrrun jrc_shelf_life_poolability.R shelf_life_pool_partial.csv`
Rationale: Dataset with equal slopes and different intercepts. Step 1 p > 0.25; Step 2 p < 0.01.
Pass criterion: exit 0; 'PARTIAL POOL' in output; Step 1 p > 0.25; Step 2 p < 0.01

TC-SHELF-POOL-005

Tests: UR-SHELF-023
Command: `jrrun jrc_shelf_life_poolability.R shelf_life_pool_poolable.csv`
Rationale: PNG output.
Pass criterion: exit 0; *_jrc_shelf_life_poolability.png found in ~/Downloads/

TC-SHELF-POOL-006

Tests: UR-SHELF-002
Command: `jrrun jrc_shelf_life_poolability.R  (no arguments)`
Rationale: Missing arguments.
Pass criterion: exit non-zero or 'Usage' in output

TC-SHELF-POOL-007

Tests: UR-SHELF-020
Command: `jrrun jrc_shelf_life_poolability.R shelf_life_pool_missing_col.csv`
Rationale: CSV missing 'batch' column.
Pass criterion: exit non-zero; 'batch' in error output

TC-SHELF-POOL-008

Tests: UR-SHELF-020
Command: `jrrun jrc_shelf_life_poolability.R shelf_life_pool_one_batch.csv`
Rationale: Only 1 batch (< 2 required).
Pass criterion: exit non-zero

TC-SHELF-POOL-009

Tests: UR-SHELF-020
Command: `jrrun jrc_shelf_life_poolability.R <2-timepoint CSV>`
Rationale: Fewer than 3 distinct time points per batch.
Pass criterion: exit non-zero

TC-SHELF-POOL-010

Tests: UR-SHELF-001

Command: `Rscript --vanilla repos/shelf_life/R/jrc_shelf_life_poolability.R shelf_life_pool_poolable.csv (RENV_PATHS_ROOT unset)`
Rationale: Direct Rscript call bypasses jrrun.
Pass criterion: exit non-zero; 'RENV_PATHS_ROOT' in output

7.5 jrc_shelf_life_extrapolate

TC-SHELF-EXT-001

Tests: UR-SHELF-024, UR-SHELF-025
Command: `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv 20`
Rationale: Valid model CSV and target within observed range.
Pass criterion: exit 0; 'Projection', 'Fitted value' in output

TC-SHELF-EXT-002

Tests: UR-SHELF-025
Command: `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv 20`
Rationale: Known numerical check (target=20): fit = 100.647 − 0.807×20 = 84.502 (±0.01). CI lower = 83.976 (±0.05) from closed-form formula with t_crit(0.975, df=13)=2.16037 (NIST t-table). References computed independently of R.
Pass criterion: exit 0; fitted value = 84.502 ± 0.01; lower 95% CI bound = 83.976 ± 0.05

TC-SHELF-EXT-003

Tests: UR-SHELF-025
Command: `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv 20`
Rationale: Lower CI > spec=80 at target=20.
Pass criterion: exit 0; 'supported' or PASS message in output

TC-SHELF-EXT-004

Tests: UR-SHELF-025
Command: `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv 30`
Rationale: Lower CI < spec=80 at target=30 (beyond shelf life).
Pass criterion: exit 0; 'NOT supported' or 'crossed' in output

TC-SHELF-EXT-005

Tests: UR-SHELF-026
Command: `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv 37`
Rationale: Target=37 is 54% beyond last_time=24 (> 50% warning threshold).
Pass criterion: exit 0; extrapolation warning ('beyond', 'wide', or '50') in output

TC-SHELF-EXT-006

Tests: UR-SHELF-026
Command: `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv 50`
Rationale: Target=50 is 108% beyond last_time=24 (> 100% hard stop).
Pass criterion: exit non-zero; '100', 'rejected', or 'speculative' in output

TC-SHELF-EXT-007

Tests: UR-SHELF-002
Command: `jrrun jrc_shelf_life_extrapolate.R  (no arguments)`
Rationale: Missing arguments.
Pass criterion: exit non-zero or 'Usage' in output

TC-SHELF-EXT-008

Tests: UR-SHELF-024
Command: `jrrun jrc_shelf_life_extrapolate.R /tmp/no_such_model.csv 20`
Rationale: Non-existent model file.

Pass criterion: exit non-zero; 'not found' in output

TC-SHELF-EXT-009

Tests: UR-SHELF-024
Command: `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_wrongscript.csv 20`
Rationale: Model file from wrong script (script field does not contain jrc_shelf_life_linear).
Pass criterion: exit non-zero; 'script' in error output

TC-SHELF-EXT-010

Tests: UR-SHELF-027
Command: `jrrun jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv -5`
Rationale: Negative target_time.
Pass criterion: exit non-zero

TC-SHELF-EXT-011

Tests: UR-SHELF-001
Command: `Rscript --vanilla repos/shelf_life/R/jrc_shelf_life_extrapolate.R shelf_life_extrapolate_model.csv 20 (RENV_PATHS_ROOT unset)`
Rationale: Direct Rscript call bypasses jrrun.
Pass criterion: exit non-zero; 'RENV_PATHS_ROOT' in output

8. Requirements Traceability Matrix

Each user requirement is covered by at least one test case.

UR ID	Title	Test Cases
UR-SHELF-001	Bypass protection	TC-SHELF-Q10-010, TC-SHELF-ARR-010, TC-SHELF-LIN-012, TC-SHELF-POOL-010, TC-SHELF-EXT-011
UR-SHELF-002	Non-zero exit on error	TC-SHELF-Q10-005..009, TC-SHELF-ARR-006..009, TC-SHELF-LIN-008..011, TC-SHELF-POOL-006..009, TC-SHELF-EXT-007, EXT-010
UR-SHELF-003	Q10 — AF computation	TC-SHELF-Q10-001, TC-SHELF-Q10-002
UR-SHELF-004	Q10 — sensitivity table	TC-SHELF-Q10-003
UR-SHELF-005	Q10 — range warning	TC-SHELF-Q10-004
UR-SHELF-006	Q10 — temp direction check	TC-SHELF-Q10-005
UR-SHELF-007	Arrhenius — AF computation	TC-SHELF-ARR-001, TC-SHELF-ARR-002
UR-SHELF-008	Arrhenius — unit flag	TC-SHELF-ARR-003
UR-SHELF-009	Arrhenius — sensitivity table	TC-SHELF-ARR-004
UR-SHELF-010	Arrhenius — Ea range warning	TC-SHELF-ARR-005
UR-SHELF-011	Arrhenius — temp/unit check	TC-SHELF-ARR-006, TC-SHELF-ARR-008
UR-SHELF-012	Linear — CSV input	TC-SHELF-LIN-001, TC-SHELF-LIN-010, TC-SHELF-LIN-011
UR-SHELF-013	Linear — Brown-Forsythe test	TC-SHELF-LIN-003, TC-SHELF-LIN-004
UR-SHELF-014	Linear — OLS + shelf life	TC-SHELF-LIN-001, TC-SHELF-LIN-002
UR-SHELF-015	Linear — --direction flag	TC-SHELF-LIN-007
UR-SHELF-016	Linear — --transform log flag	TC-SHELF-LIN-013, TC-SHELF-LIN-014, TC-SHELF-LIN-015, TC-SHELF-LIN-016
UR-SHELF-017	Linear — PNG output	TC-SHELF-LIN-005
UR-SHELF-018	Linear — model CSV output	TC-SHELF-LIN-006, TC-SHELF-LIN-016
UR-SHELF-019	Linear — reject below spec at t=0	TC-SHELF-LIN-008
UR-SHELF-020	Poolability — CSV input	TC-SHELF-POOL-001, TC-SHELF-POOL-007, TC-SHELF-POOL-008, TC-SHELF-POOL-009
UR-SHELF-021	Poolability — ANCOVA	TC-SHELF-POOL-001, TC-SHELF-POOL-002, TC-SHELF-POOL-003, TC-SHELF-POOL-004
UR-SHELF-022	Poolability — pooling decision	TC-SHELF-POOL-002, TC-SHELF-POOL-003, TC-SHELF-POOL-004
UR-SHELF-023	Poolability — PNG output	TC-SHELF-POOL-005

UR-SHELF-024	Extrapolate — model CSV + verify	TC-SHELF-EXT-001, TC-SHELF-EXT-008, TC-SHELF-EXT-009
UR-SHELF-025	Extrapolate — CI projection	TC-SHELF-EXT-001, TC-SHELF-EXT-002, TC-SHELF-EXT-003, TC-SHELF-EXT-004
UR-SHELF-026	Extrapolate — ICH Q1E limits	TC-SHELF-EXT-005, TC-SHELF-EXT-006
UR-SHELF-027	Extrapolate — negative time check	TC-SHELF-EXT-010

9. Acceptance Criteria

The Shelf Life & Degradation Analysis module OQ is considered PASS when all of the following conditions are satisfied:

- All 57 automated test cases execute with the result PASSED.
- Zero test cases result in FAILED or ERROR.
- The OQ evidence file is generated and preserved at ~/.jrscript/<PROJECT_ID>/validation/shelf_life_oq_execution_<timestamp>.txt.
- All test cases are executed using the validated jrrun entry-point (bypass protection verified for all 5 scripts).
- No open deviations remain unresolved at the time of approval.

Any test failure shall be investigated, documented as a deviation, resolved, and the affected test(s) re-executed. The OQ may not be closed until all deviations are resolved.